Fiscal Year:	FY 2013	**Task Last Updated:**	FY 02/05/2013
PI Name:	Baulch, Janet E. Ph.D.		
Project Title:	High LET Radiation Induced Carcinogenesis: Epigenetic Mechanisms in Mouse Models of Human Cancers		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92697	**Congressional District:**	45
Comments:	NOTE: PI previously at University of Maryland; moved to UCA Irvine in summer 2013 (Ed., 11/11/13)		
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Radiobiology NNJ11ZSA001N
Start Date:	03/26/2012	**End Date:**	07/09/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/9/2013 due to PI move to UCA, Irvine (Ed., 11/11/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX12AE79G		
Performance Goal No.:			
Performance Goal Text:			
	This is a collaborative project with other NASA investigators who are characterizing mouse models of human lung and colon cancers. Data from both our laboratory and the literature suggest that epigenetics play a significant role in carcinogenic processes. By describing the changes in DNA methylation and miR regulation of gene expression for these mice, we will provide NASA with mechanistic data regarding carcinogenic initiation and progression. By using normal, cancer, and metastatic cancer cell lines we will obtain information regarding epigenetic changes that occur during carcinogenesis based on stage. For example, loss of control of gene expression for extracellular matrix (ECM) remodeling genes may foster metastatic behavior in colon cancer. For this reason we will include ECM gene promoter DNA methylation and gene expression among the loci evaluated in this study to the hypothesis that dis-regulation of epigenetic marks and gene expression changes with disease progression. Similarly, the work of other NASA		

Task Description:	investigators suggests that we will see epigenetically regulated differences among cell lines in expression of inflammatory response pathway gene regulation among lung cell lines. These experiments will provide information regarding radiation responses of these mouse tissues and human cell lines based on the LET of the radiation used. Abnormal epigenetic silencing of genes can occur at a variety of times during tumor progression. If so, does radiation exposure push normal cells to a cancer phenotype and cancer cells to a metastatic phenotype? This question is of importance to NASA since astronauts are of an age where they likely go into missions with precursor lesions. It is important to understand whether their occupational radiation exposures lead to increased cancer risk. Tasks: 1. Evaluate normal and cancer cell lines from lung and colon for responses to low dose high LET radiation exposure 2. Establish epigenetic profiles for in vivo radiation induced cancers in mouse models of human lung and colon cancers. [Editor's note 11/11/2013: See grant NNX13AK69G with the same title for subsequent reports, due to PI move to University of California, Irvine, in summer 2013]
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Reducing the uncertainties associated with cancer risk models for radiation exposures involved in space flight is of considerable importance in light of long-term occupation of the International Space Station (ISS) and other future manned missions by astronauts. A greater understanding these carcinogenic mechanisms also benefits Earth-bound populations. Colon cancer is the third most common cancer. Precancerous lesions in the GI tract are found in approximately 1 out of every 10 people by age 40 and this incidence increases significantly with age. In addition to being the largest target for radiation effects, lung cancer is the most prevalent form of cancer in men and women. Each of these cancers has hallmark mutations associated with them. On top of these genetic changes, our understanding of carcinogenesis is complicated by epigenetic alterations to the genome. These epigenetic changes result in misregulation of gene expression for pathways such as cell cycle regulation, DNA repair, apoptosis, and extracellular matrix remodeling. A better understanding of these epigenetic mechanisms may lead to improvements in clinical treatments for lung, colon, and other cancers.
Task Progress:	The plan for this project is to evaluate epigenetic mechanisms in mouse tissues from other NASA investigators involved in the Lung and Colon cancer NSCORs. In this first year, we have begun liaising with other NASA investigators to establish which experimental groups (genotypes, time points, doses, radiations) will have the highest priority and provide the most useful information. We have also established what is required with regards to tissue collection and preservation. We have obtained the majority of the cell lines that will be used in this study and began propagating them and archiving the appropriate cells stocks. In conjunction with another of our NASA funded projects, we have developed the pyrosequencing assays that will be used in the analysis of DNA methylation. During this process we have confirmed a tissue-specific effect of irradiation on DNA methylation in the lung of normal male mice. We have also applied for and received approval for the first NSRL campaign related to this project, NSRL13A. During this run we will irradiate lung cell lines that are both normal and cancerous using 600 MeV/μ 56Fe ions. These cell lines will then be propagated for analysis of delayed changes in DNA methylation of gene promoters and repeat elements within the genome, and mRNA and miR expression. We also anticipate receiving our first set of normal tissue and tumor samples from collaborators within the coming 1-2 months. [Editor's note 11/11/2013: See grant NNX13AK69G with the same title for subsequent progress, due to PI move to University of California, Irvine, in summer 2013]
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